Starter

Look at the word equation below:

Potassium chloride + Fluorine -> Potassium fluoride + Chlorine

1. What has happened to the **chlorine** and **fluorine** in this reaction?
2. Why do you think **fluorine** was able to *kick* **chlorine** out of its compound?

   (Hint: Think about reactivity down Group 7!)

Starter

Look at the word equation below:

Potassium chloride + Fluorine -> Potassium fluoride + Chlorine

1. What has happened to the **chlorine** and **fluorine** in this reaction?

   Fluorine displaces chlorine

2. Why do you think **fluorine** was able to *kick* **chlorine** out of its compound?

   Fluorine is more reactive than chlorine

Trends in groups within the Periodic Table:

Group 0 elements

Learning Objectives

- To interpret data to describe patterns in properties of the Group 0 elements.
- To use patterns to predict properties of Group 0 elements.

Brain Teaser

What does the word ‘noble’ mean?

Group 0 elements		
Meeting	Exceeding	Mastering
I can list the group 0 elements and describe their properties as part of the noble gases.	I can explain the group 0 elements are inert due to their valence shell electrons.	I can explain the trends in the noble gases as I read up/down the group.

Success criteria

Lesson Title	Meeting	Exceeding	Mastering
Atomic Structure	☐ I can name the subatomic particles and their charges	☐ I can explain how electrons are configured on different atoms	☐ I can use the periodic table to draw the electron configuration for multiple atoms.
Group 1 - The Alkali Metals	☐ I can list the group 1 elements and describe their properties as part of the alkali metals.	☐ I can explain the group 1 elements reactivity in relation to their valence shell electrons	☐ I can explain the trends in the alkali metals as I read up/down the group
Group 7 - The Halogens	☐ I can list the group 7 elements and describe their properties as part of the halogens.	☐ I can explain the group 7 elements reactivity in relation to their valence shell electrons	☐ I can explain the trends in the halogens as I read up/down the group
Group 0 - The Noble Gases	☐ I can list the group 0 elements and describe their properties as part of the noble gases.	☐ I can explain the group 0 elements are inert due to their valence shell electrons	☐ I can explain the trends in the noble gases as I read up/down the group
Ionic Bonding	☐ I can describe ionic bonding as the combining of a metal atom with a nonmetal atom	☐ I can explain why metals lose electrons to because positive ions and nonmetals gain electrons to become negative ions	☐ I can use the periodic table to draw accurate dot and cross diagrams to demonstrate ionic bonds
Covalent Bonding	☐ I can describe covalent bonding as the combining of two or more nonmetal atoms	☐ I can explain how atoms will share valence electrons and form covalent molecules	☐ I can use the periodic table to draw accurate dot and cross diagrams of single, double or triple covalent bonds

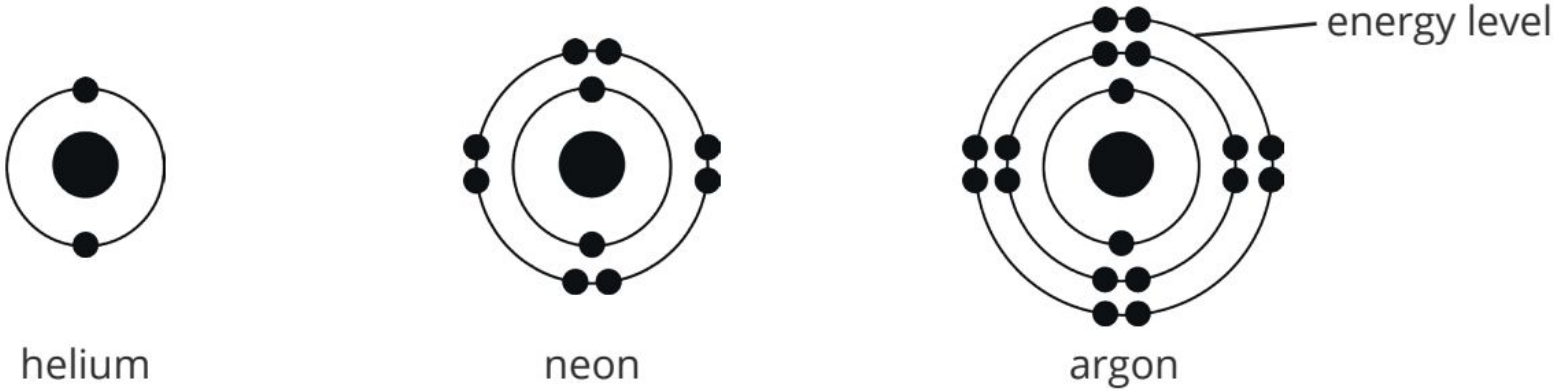
Group 0
The Noble Gases

They are inert (unreactive) and do not form compounds

All group 0 Element has a Full outer shell electrons

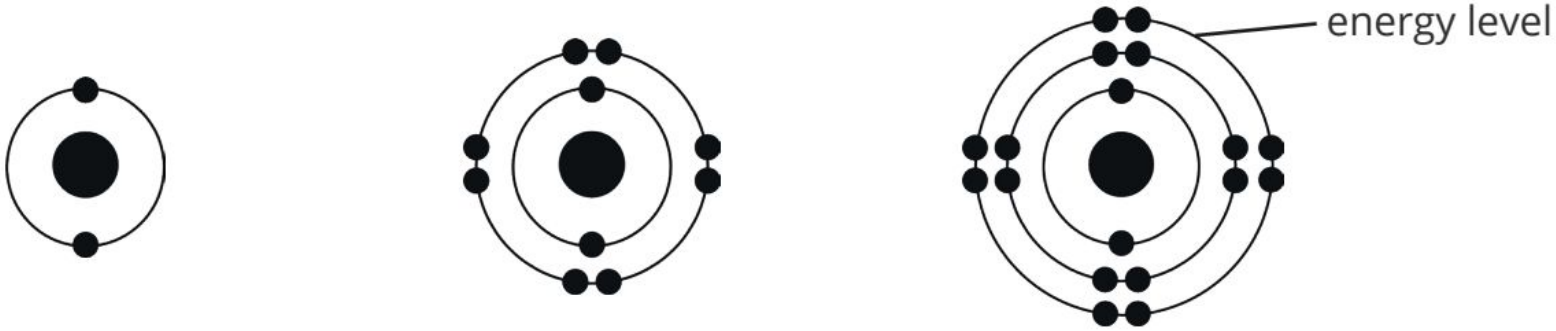


The diagrams show the first three Group 0 elements. Each ring represents an energy level.

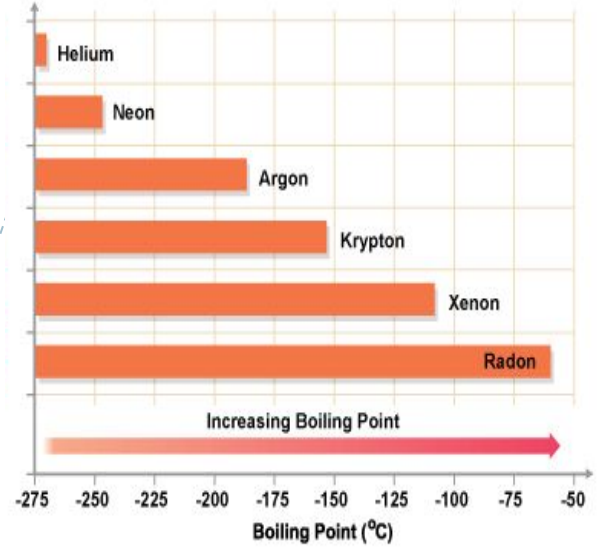


Explain the trend in the size of the atoms of the noble gases.

The diagrams show the first three Group 0 elements. Each ring represents an energy level.



As you move **down the group**, each atom has an **extra energy level/shell/layer**. This means that the **radius/size of the atoms increases** as you move down the group.

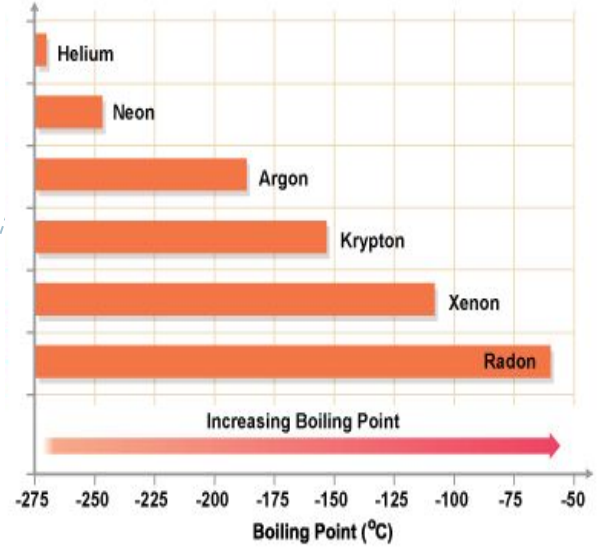


Element	Boiling Point (°C)
helium	-269
neon	-246
argon	-186
krypton	
xenon	-108
radon	-62

1. Explain why the Group 0 elements are gases at room temperature.

2. Describe the trend in the boiling point of the Group 0 elements.

3. Predict the boiling point of krypton.



Element	Boiling Point (°C)
helium	-269
neon	-246
argon	-186
krypton	
xenon	-108
radon	-62

1. Their boiling points are all lower than room temperature (around 23°C).

2. The boiling point increases as you move down the group.

3. Any value between -109 and -185 °C.

Group 8 - The Noble Gases

THE GROUP 8 ELEMENTS ARE ALL ODOURLESS, COLOURLESS, MONOATOMIC GASES WITH A VERY LOW CHEMICAL REACTIVITY

2 He Helium

10 Ne Neon

18 Ar Argon

36 Kr Krypton

54 Xe Xenon

86 Rn Radon

BOILING POINTS



THE NOBLE GASES ARE ALL **MONOATOMIC** AND **RARELY FORM COMPOUNDS**



ALL OF THE GROUP 8 ELEMENTS HAVE A *FULL OUTER SHELL* OF ELECTRONS

THE GROUP 8 ELEMENTS ARE **VERY UNREACTIVE** AS THEY ALREADY HAVE A *FULL VALENCE ELECTRON SHELL*

-269°C **HELIUM** HAS THE LOWEST BOILING POINT OF ALL ELEMENTS IN THE PERIODIC TABLE

THE COLOURS IN 'NEON' SIGNS ARE CAUSED BY IONISED NOBLE GASES

He Ne Ar Kr Xe

1 H 75%

2 He 23%

3 O 1%

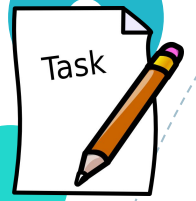
HELIUM *IS THE* **SECOND MOST ABUNDANT ELEMENT** IN THE UNIVERSE

RADON IS **RADIOACTIVE** AND CAN ENTER HOMES THROUGH BASEMENTS AFTER ***RADIOACTIVE DECAY OF ROCKS*** BELOW THE EARTH

^{222}Rn

USES OF THE NOBLE GASES

HELIUM	NEON	ARGON	KRYPTON	XENON
BALLOONS	NEON LIGHTS	MEDICAL LASERS	CAMERA FLASHES	MEDICAL IMAGING
	REFRIGERANT	LIGHT BULBS		



Create a ‘knowledge organiser’ covering the key content on Group 1, 7 and 0.

It should contain the following.

a)The **physical properties** of the groups

(Trends for **boiling point and melting point.** Are they solid, liquid or gas at room temperature?)

b) The **reactivity as it goes down** the Group

c) The **trends in the atomic size**

d) **Special characteristics** – colour, the reaction with…

You may use

- **A4 (1 pages)**

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